

Customer Data Analysis

Analysis Report | 2021 – 2023

Total Customers	99,457		
Total Revenue	68.55 Million TL		
Time Period	2021 – 2023		
Location	Istanbul, Turkey (10 Malls)		
Product Categories	Clothing, Cosmetics, Food & Beverage, Toys, Shoes, Souvenir, Technology, Books		
Payment Methods	Cash, Credit Card, Debit Card		
Report Prepared By	Bhuvaneshwaran K		
Visualizations	Microsoft power BI		
299K	68.55M	8	3
Transactions 2021–2023	Total Revenue Turkish Liras	Product Categories Across 10 malls	Payment Methods Cash · Credit · Debit

01 Executive Summary

This report analyses shopping transactions across 10 major Istanbul malls from 2021 to 2023. Nearly 300,000 transactions spanning eight product categories, three age groups, two genders, and three payment methods provide a robust foundation for understanding consumer behaviour in one of Turkey's most active retail markets.

Three structural themes define the dataset: **(1)** female customers account for nearly 60% of all transactions and revenue; **(2)** the 40-and-above age cohort drives more than half of all volume and revenue; and **(3)** cash remains the dominant payment method despite global digitalisation of retail payments. Each theme carries distinct strategic implications covered in the sections below.

02 Gender Analysis

Shopping Distribution

Female customers account for **59.81% of all transactions (179K)** versus 40.19% for males (120K). This near 60/40 split holds consistently across all three years, all eight product categories, and all three payment methods — confirming it is a structural demographic pattern, not a product-preference effect.

Female 59.81% (179K)

Male 40.19% (120K)

Revenue Generated

40.93M	27.62M	68.55M
Female Revenue 59.71% of total	Male Revenue 40.29% of total	Combined Total 2021–2023

Gender Distribution Across Categories

Every category sits within one percentage point of the overall 60/40 ratio, ruling out product preference as the driver of the gender gap. The gap is demographic.

Category	Female	Male	Total	Female %
Clothing	62k	42k	104k	59.6%
Cosmetics	27k	18k	45k	60.0%
Food & Beverage	26k	18k	44k	59.1%
Shoes	18k	12k	30k	60.0%
Toys	18k	12k	30k	60.0%
Books	9k	6k	15k	60.0%
Technology	9k	6k	15k	60.0%
Souvenir	9k	6k	15k	60.0%
Total	179k	120k	299k	59.9%

Key Gender Insight

The 60/40 female-to-male ratio is structural, not preference-driven. Closing the gap requires demand-generation at the awareness and acquisition stage, not product assortment changes.

03 Age Group Analysis

Shopping Volume by Age

The 40+ cohort dominates at **55K shoppers** — nearly double the 26–40 group (29K) and more than triple the 18–25 segment (15K). This reflects both Istanbul's mall-going demographic profile and the greater disposable income of older groups.

40 and above		55K
26 to 40		29K
18 to 25		15K

Revenue by Age Group

38.11M 40+ Revenue 55.6% of total	19.99M 26–40 Revenue 29.2% of total	10.44M 18–25 Revenue 15.2% of total
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Average revenue per transaction is remarkably flat at approximately **693–697 TL** across all three groups, confirming that purchasing behaviour is consistent across the age spectrum. The 40+ group's dominance is purely a volume effect.

Key Age Insight

The 18–25 segment represents only 15% of volume despite being a large population cohort. This under-representation signals an acquisition gap — young consumers are not visiting these malls at rates proportional to their population share.

04 Product Category Distribution

Category Volume Overview

Clothing dominates at **104K units** — more than double the next largest category (Cosmetics at 45K). Food & Beverage (44K), Shoes (30K), and Toys (30K) form a mid-tier. Books, Technology, and Souvenir each account for only 15K units, representing a long-tail with significant growth potential.

Clothing		104K
Cosmetics		45K
Food & Beverage		44K
Shoes		30K
Toys		30K
Books		15K
Technology		15K
Souvenir		15K

Category Concentration Risk

Clothing represents approximately **34.8%** of all units sold. This concentration creates revenue vulnerability — a shift in fashion trends, competitive pressure from e-commerce, or macroeconomic stress on discretionary spending could materially impact overall mall performance. The three smallest categories together account for only 15% of volume, indicating significant room for category development.

Cross-Category Observation

All 8 categories show an identical ~60/40 female-to-male purchase split. This uniformity across categories as different as Technology and Cosmetics strongly implies the gender ratio is set at mall entry, not at store level.

05 Payment Method Analysis

Payment Method Distribution

Cash		44.69%
Credit Card		35.12%
Debit Card		20.19%

Payment Method by Gender & Revenue

Payment Method	Transactions (K)	Share %	Female (M TL)	Male (M TL)
Cash	44.45	44.69%	18.4	12.3
Credit Card	34.93	35.12%	14.4	9.65
Debit Card	20.08	20.19%	8.13	5.67
Total	99.46	100.0%	40.93	27.62

Implications of Cash Dominance

- Cash transactions yield zero customer-level data, limiting personalisation capability.
- Cash handling incurs higher operational costs — counting, security, and reconciliation.
- The cash preference likely correlates with the 40+ age-dominant base and may be a cultural/generational pattern that will naturally decline as younger, card-preferring cohorts grow into higher-spending brackets.

06 Cross-Dimensional Insights

Gender × Age

The consistent 60/40 female-majority is not age-cohort specific — it appears equally in the 18–25 and 40+ groups, confirming the pattern is set at the population level.

Age × Payment

Cash dominance likely correlates with the 40+ majority. As younger, card-preferring shoppers grow their spending share, the payment mix will shift naturally over time.

Category × Age

Technology (15K) being one of the smallest categories despite the presence of 18–25 shoppers suggests limited stock, poor placement, or pricing misalignment for the age group most likely to buy tech products.

Revenue per Transaction

At approximately 695 TL per transaction across all genders and age groups, average spend is uniformly flat. Revenue growth must therefore come from increased visit frequency or new customer acquisition — not from upselling existing customers.

07 Strategic Recommendations

PRIORITY**Win the 18–25 Segment**

This cohort is structurally under-indexed at only 15% of volume.

- Introduce BNPL (Buy Now Pay Later) payment options to reduce price sensitivity.
- Run TikTok/Instagram-integrated campaigns targeting Gen Z shoppers.
- Launch student discount programmes across Clothing, Technology, and Books.
- A 5pp share gain represents approximately 15K new annual transactions.

PRIORITY**Convert Cash Payers to Card**

Nearly 45% of transactions are in cash, yielding zero customer-level data.

- Offer loyalty point multipliers or instant cashback for card payments.
- Targeting 10% conversion of cash transactions generates ~4,400 new trackable records per year.
- Invest in contactless POS terminals at high-volume Clothing and Food & Beverage counters.

PRIORITY**Build Male-Targeted Experiences**

Technology, Toys, and Shoes are most likely to resonate with male shoppers.

- Activate sports tie-ins, tech launches, and sneaker drop events.
- Even shifting to a 45/55 male-to-female ratio adds ~15K male transactions annually.

PRIORITY**Develop Under-Performing Categories**

Books, Technology, and Souvenir each hold only 15K units (5% of total volume).

- Introduce cross-category bundles — e.g., Clothing + Technology gift sets.
- Co-market with anchor Clothing tenants to redirect footfall to smaller stores.
- Curated in-mall events and pop-ups can raise category awareness at low cost.

LONG-TERM**Premium Offering for 40+ Loyalty Base**

The 40+ group is the most valuable segment and likely the most loyal.

- Introduce premium membership tiers with exclusive parking and early-access sales.
- Personal shopping assistance can increase visit frequency and average spend.

LONG-TERM**Data Infrastructure Investment**

With 44.7% cash transactions, the malls operate with significantly degraded customer intelligence.

- Implement loyalty sign-ups, QR receipts, and WiFi registration to capture cash-payer data.
- True personalisation is only possible once a complete customer profile can be built.

08 Conclusions

Female dominance is structural.

The 60/40 female-to-male split is consistent across every category, age group, and payment method. It will not change through product assortment. Only targeted male acquisition campaigns can close the gap.

The 40+ cohort is the revenue backbone.

Accounting for 55% of transactions and revenue, this group should be rewarded with premium loyalty benefits. However, over-reliance on a single demographic is a long-term risk.

Growth requires new visitors, not higher spend.

At a flat ~695 TL per transaction, basket inflation is not a viable lever. Every meaningful revenue target must be achieved through footfall growth — particularly among younger and male shoppers.

Cash is both a symptom and an obstacle.

High cash usage reflects the 40+ demographic. It also suppresses customer data. Digitising payments is the single action that simultaneously reduces cost, improves data, and positions the malls for the next generation of shoppers.

Immediate priorities are clear.

Three initiatives address the three most underserved segments: (1) incentivise card payments to unlock data, (2) activate the 18–25 segment through digital channels, and (3) deploy male-targeted campaigns in Technology, Toys, and Shoes.